HACKATHON 2025 · BANGALORE TRAFFIC · AI + CV

TrafficEye AI

Intelligent Traffic Violation Detection & Enforcement System

An AI-powered computer vision platform built specifically for **Bengaluru Traffic Police (BTP)** — transforming passive surveillance into an active, intelligent enforcement ecosystem that detects violations in real-time, generates legal evidence, and delivers predictive enforcement intelligence.

82.86L	893	4,784	■80.9Cr
Traffic Cases Registered (2024)	Road Deaths in 2024	Road Crashes in 2024	Fines Collected (2024)

Source: Bengaluru Traffic Police (BTP) Annual Report 2024 · Deccan Herald

Tech Stack	YOLOv11 · DeepSORT · PaddleOCR · FastAPI · React · PostgreSQL · Docker
Domain	Computer Vision · Traffic Enforcement · Predictive AI · Smart City
Target	Bengaluru Traffic Police (BTP) · ITMS Integration · e-Challan System

1. The Bengaluru Traffic Crisis

Bengaluru — India's Silicon Valley — is simultaneously the nation's most congested city and one of its deadliest for road users. Despite deploying India's most advanced Intelligent Traffic Management System (ITMS), the scale of violations overwhelms human enforcement capacity.

METRIC	FIGURE	IMPACT
Total traffic cases (2024)	**82.86 lakh**	~22,700 violations per day
Contactless enforcement (2025)	**88% of all cases**	Camera-driven, nearly zero manpower
Road deaths (2024)	**893 fatalities**	Highest in Karnataka
Fatal crashes (2024)	**871 crashes**	Down only 4% from 2023
Pedestrian + biker deaths	**91% of all fatalities**	Critical vulnerability gap
Rash driving cases (2025)	**+82% surge YoY**	Enforcement gaps remain
Wrong-direction driving	**6,872 BNS cases (2025)**	Up from 3,774 in 2024
Drunk driving caught	**23,574 drivers (2024)**	12.54 lakh vehicles checked
Fines collected	**■80.9 crore (2024)**	Under-captures 94% violations
Single station violations	**1.02 lakh (Sadashivanagar)**	Until May 2024 alone

Sources: BTP Annual Report 2024 · Deccan Herald Jan 2025 · Deccan Herald May 2025

1.1 Gaps in Existing BTP-ITMS Infrastructure

Bengaluru's ITMS (launched Dec 2022) covers **50 junctions with 250 ANPR + 80 RLVD cameras**. While award-winning (ET Infra Leadership Award 2025), critical gaps remain:

- ✗ **No helmet / seatbelt AI:** ITMS cameras primarily detect stop-line and red-light violations. Rider safety violations require separate CV models.
- ✗ **No triple-riding detection:** Motorcycle occupant counting is absent from current ITMS pipeline.
- ✗ **Zero repeat-offender intelligence:** BTP has no automated cross-station offender profiling system.
- ✗ **No predictive deployment tool:** Resources are reactively deployed rather than data-driven.
- ✗ **Coverage blind spots:** Hand-held FTVR devices are used where no ITMS cameras exist — labour-intensive.
- ✗ **No violation impact scoring:** All violations are treated equally; high-risk zones are not prioritised.

2. TrafficEye AI — Proposed Solution

TrafficEye AI is not another CCTV analytics wrapper. It is a **full-stack intelligent enforcement platform** designed as a plug-in upgrade layer on top of BTP-ITMS or any camera feed — adding AI-driven violation detection, ANPR, evidence packaging, and predictive enforcement intelligence.

2.1 Core Innovation Pillars

Violation Impact Score (VIS)

Each detected v
detection, locati
see HIGH → M

Repeat Offender Graph

All violations are
vehicles with 3+
a gap BTP itsel

Predictive Enforcement Intelligence

ML models (trai
hour, day, and w
junction 6–9 PM

One-click Legal Evidence Package

Every violation
ANPR plate tex

2.2 Violations Detected (Bengaluru-Specific Focus)

VIOLATION TYPE	CV APPROACH	BTP PRIORITY	VIS LEVEL
Helmet Non-Compliance	YOLOv11 person + helmet classifier	CRITICAL — 91% biker fatalities	HIGH
Triple Riding	Occupant counter on motorcycle bbox	HIGH — common Bangalore violation	HIGH
Seatbelt Non-Compliance	Keypoint analysis on driver region	MEDIUM	MEDIUM
Red-Light Violation	Traffic light state + vehicle motion	CRITICAL — major fatality cause	HIGH
Stop-Line Violation	Line-crossing detection (DeepSORT)	HIGH	MEDIUM
Wrong-Side Driving	Direction vector vs. road flow	CRITICAL — BNS cases +82% in 2025	HIGH
Illegal Parking	Zone mask + vehicle dwell time	HIGH — HAL, Bellandur, HSR hotspots	MEDIUM
Mobile Phone Use	Hand-to-ear pose detection	HIGH — 83% of Bengalureans guilty	MEDIUM

3. System Architecture

The pipeline is a modular, containerised system. Each stage is independently scalable and plugs into BTP's existing ITMS camera feeds via RTSP streams or image batch uploads.

INGEST	**Camera Feed / Image Upload**	RTSP stream fr H.264 streams
PREPROCESS	**Image Enhancement**	CLAHE low-ligh normalisation
DETECT	**Vehicle & Road User Detection**	YOLOv11 objec confidence · De
CLASSIFY	**Violation Detection Engine**	Per-class violati Signal state det
OCR	**License Plate Recognition (ANPR)**	YOLOv11 plate (IND-VAAHAN r
SCORE	**Violation Impact Score**	Rule-based + M accident index
EVIDENCE	**Evidence Package Generator**	Annotated imag Tamper-evident
ANALYTICS	**Dashboard & Predictive Intelligence**	React real-time prediction, peak

4. Technology Stack

COMPUTER VISION

- YOLOv11 (Ultralytics)
- OpenCV 4.x
- DeepSORT tracker
- Albumentations (augmentation)

OCR / ANPR

- PaddleOCR (primary)
- EasyOCR (fallback)
- Indian plate regex parser
- VAAHAN number format validator

BACKEND API

- FastAPI (async)
- Celery + Redis (task queue)
- SQLAlchemy ORM
- Pydantic v2 schemas

DATABASE

- PostgreSQL (violations DB)
- TimescaleDB (time-series analytics)
- Redis (caching + pub-sub)
- S3-compatible object storage

FRONTEND

- React 18 + Vite
- Tailwind CSS
- Recharts (analytics)
- Leaflet.js (heatmaps)
- Socket.IO (live feed)

ML / ANALYTICS

- scikit-learn (VIS model)
- Prophet / LSTM (forecasting)
- Pandas + NumPy
- Jupyter for model dev

INFRA / DEVOPS

- Docker + Docker Compose
- NGINX reverse proxy
- GitHub Actions CI/CD
- Prometheus + Grafana

5. GitHub Reference Repositories

These production-grade open-source repositories directly underpin TrafficEye AI's implementation. Each is battle-tested, actively maintained, and architecturally relevant.

Core Detection Framework

◆ ultralytics/ultralytics	■ 40k+	m/ultralytics/ultra
◆ nwojke/deep_sort	■ 8k+	m/nwojke/deep_sort

Traffic Violation Systems

◆ anuragparashar26/traffic-management	Full-stack	m/anuragparashar26/
◆ festorean/Smart-Traffic-Monitoring-System-using-AI	Detection	m/festorean/Smart-T
◆ Juliowiwiwiwi/Traffic-Management-System-With-Ai	Full-stack	m/Juliowiwiwiwi/Tra
◆ ErikElcsics/Speed-Limit-Violation-Detection-System	Detection	m/ErikElcsics/Speed

Indian ANPR

◆ lavanyashree2805/yolov8-license-plate-india	India-specific	m/lavanyashree2805/
◆ Muhammad-Zeerak-Khan/Automatic-License-Plate-Recognition-using-YOLOv8	ANPR	m/Muhammad-Zeerak-K
◆ ikigai-aa/Automatic-License-Plate-Recognition	Edge AI	m/ikigai-aa/Automat

OCR & Preprocessing

◆ PaddlePaddle/PaddleOCR	■ 45k+	m/PaddlePaddle/Padd
◆ JaidedAI/EasyOCR	■ 23k+	m/JaidedAI/EasyOCR

Datasets & Benchmarks

◆ AI City Challenge Track 5	Dataset	ychallenge.org/
◆ Roboflow Traffic Datasets	Dataset	roboflow.com/search
◆ OpenCity Bengaluru	BTP Data	city.in/dataset/ben

6. Hackathon Implementation Roadmap

A realistic 2-day hackathon sprint plan to build a functional demo.

PHASE 1 · Day 1, Morning (0–4 hrs)

✓ Set up FastAPI skeleton + PostgreSQL schema (violations, vehicles, offenders tables)
✓ Integrate YOLOv11 pre-trained COCO weights for vehicle + person detection
✓ Run inference on sample Bengaluru traffic images / BTP ITMS screenshot frames
✓ Build ANPR pipeline: YOLOv11 plate detector → PaddleOCR → Indian regex parser

PHASE 2 · Day 1, Afternoon (4–8 hrs)

✓ Add helmet, seatbelt, triple-riding violation classifiers (fine-tune or rule-based)
✓ Implement DeepSORT tracker for persistent vehicle IDs across frames
✓ Build Violation Impact Score engine (weighted scoring function)
✓ Store violation records + annotated image snapshots to PostgreSQL + local storage

PHASE 3 · Day 2, Morning (8–14 hrs)

✓ Build React dashboard: live violations feed, KPI cards, area-wise heatmap
✓ Add repeat-offender detection logic (cross-reference plate DB)
✓ Implement simple predictive model: violation frequency by hour/location (Pandas + sklearn)
✓ Evidence PDF generator: annotated image + metadata bundle per violation

PHASE 4 · Day 2, Afternoon (14–20 hrs)

✓ End-to-end demo pipeline: sample video → detections → dashboard → evidence PDF
✓ Docker Compose packaging (frontend + backend + DB in one command)
✓ Demo video recording + pitch deck alignment
✓ Stretch: e-Challan API mock endpoint + mobile number SMS alert prototype

7. Expected Outcomes & Bangalore Impact

OUTCOME	METRIC	BANGALORE RELEVANCE
Automated violation detection	**7 violation types, real-time**	Replaces 600+ manual officers' check points
ANPR accuracy	**>90% on Indian plates**	Handles non-standard Bangalore plates
Evidence generation speed	**<2 sec per violation**	vs. 30+ min manual process at BTP
Repeat offender identification	**Auto-profiling from Day 1**	BTP currently has no such system
Predictive deployment	**Hotspot forecasts 48hr ahead**	Data-driven BTP resource allocation
Coverage expansion	**Any camera feed, not just ITMS**	Covers 94% of Bangalore junctions
Cost reduction	**80% less manual inspection**	■80.9Cr fines → ■250Cr potential

8. Future Scope — Beyond the Hackathon

e-Challan API Integration		**Real-time RTSP Stream Processing**	Process live ITMS simultaneously usi
Edge AI on Camera Hardware		**Accident Risk Prediction**	Combine violation features to predict
Emergency Vehicle Prioritisation		**Multi-city ITMS Integration**	Extend to Mysuru corridors proposed

> *TrafficEye AI is built for Bengaluru — using BTP's own data, targeting BTP's own gaps, and designed to slot directly into the city's existing ITMS infrastructure. Every statistic in this proposal came from official BTP reports and Karnataka government data. This isn't a generic traffic project — it is a Bengaluru enforcement upgrade.*